FORM TP 2014045

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C A R I B B E A N E X A M I N A T I O N S C O U N C I L

CARIBBEAN SECONDARY EDUCATION CERTIFICATE® EXAMINATION

BIOLOGY

Paper 02 – General Proficiency

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in two sections. Answer ALL questions.
2. For Section A, write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
3. For Section B, write your answers in the spaces provided at the end of each question, in this booklet.
4. Where appropriate, answers should be illustrated by diagrams.
5. If you need to re-write any answer and there is not enough space to do so on the original page, you must request extra lined pages from the invigilator. **Remember to draw a line through your original answer and correctly number your new answer in the box provided**.
6. **If you use extra pages you MUST write your registration number and question number clearly in the boxes provided at the top of EVERY extra page.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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SECTION A

Answer ALL questions.

Write your answers in the spaces provided in this booklet.

1. (a) Figure 1 shows drawings of different types of fish collected by marine biologists who are studying the distribution of organisms living in a marine ecosystem.

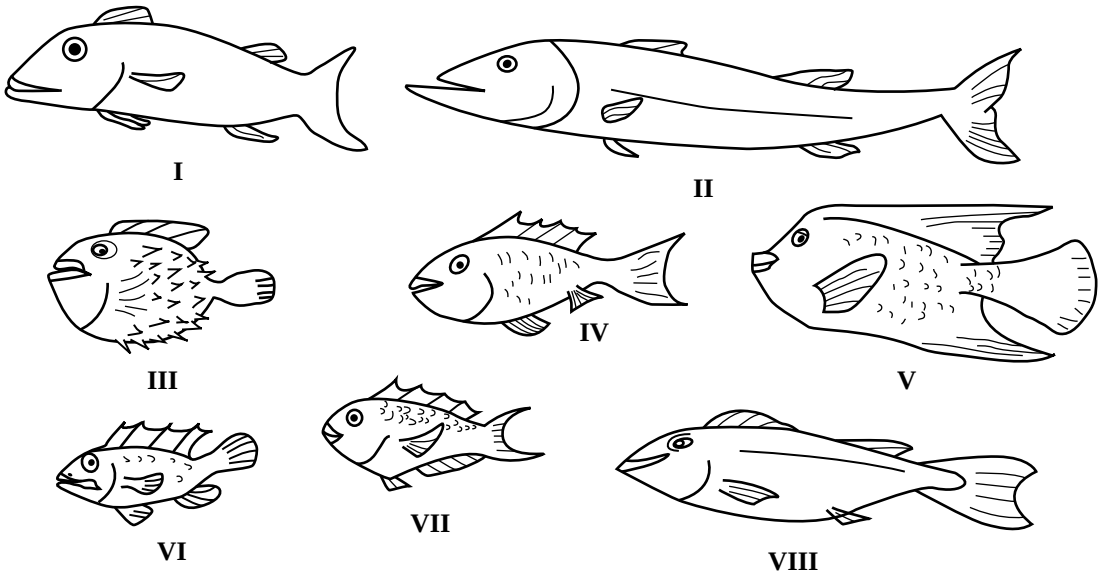

Figure 1. Fish collected from a marine ecosystem

(i) Construct a table in the space below to classify the fish in Figure 1 into groups using TWO visible features **other than size.**

(3 marks)

(ii) A gill from one of the fish is shown in Figure 2. Annotate Figure 2 to illustrate TWO characteristics that make the gill suitable for gaseous exchange.

(2 marks)

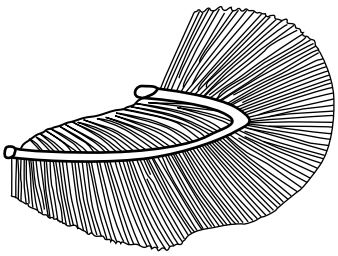

Figure 2. Gill of a fish

(iii) State TWO reasons why gaseous exchange is important in living organisms.

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(2 marks)

NOTHING HAS BEEN OMITTED.

(b) Figure 3 shows a food web that the biologists constructed to describe the feeding relationships observed in this marine ecosystem.

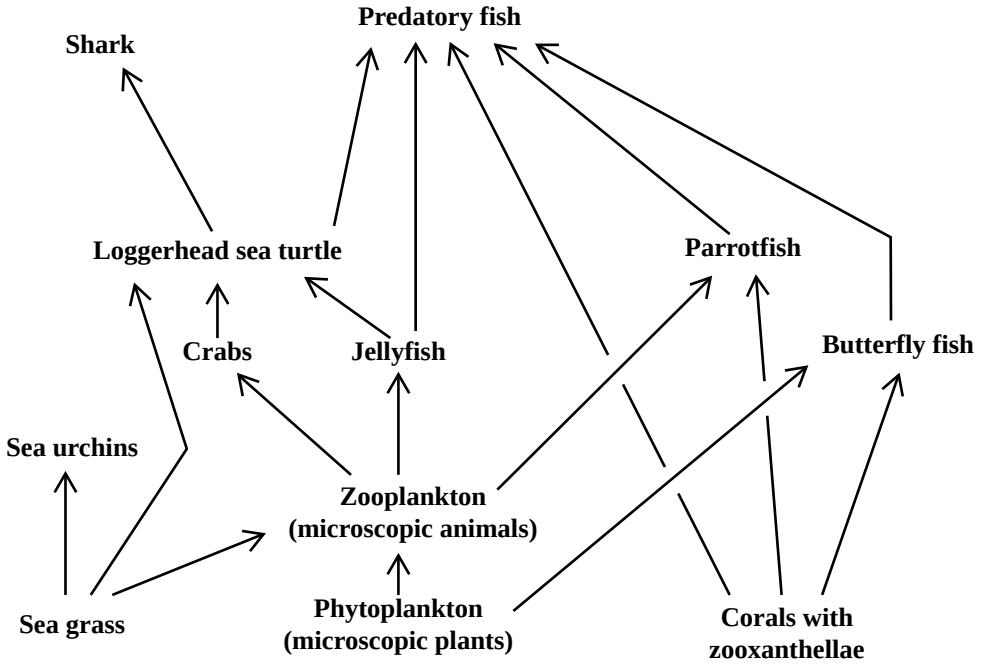

Figure 3. Food web in the marine ecosystem

(i) Identify TWO producers in the food web in Figure 3.

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(2 marks)

(ii) State TWO characteristics of producers in a food web.

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(2 marks)

(iii) Corals are found in shallow waters and contain microscopic algae known as zooxanthellae which contain chloroplasts. The corals provide the algae with carbon dioxide, shelter and protection, while the algae provide the corals with nutrients and oxygen.

Identify the type of feeding relationship between the coral and the zooxanthellae and give a reason for your answer.

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(2 marks)

(c) At six-month intervals over a three-year period, marine biologists collected data on the estimated population size of parrotfish and a predatory fish. Table 1 presents this data.

TABLE 1: ESTIMATED NUMBER OF PARROTFISH AND PREDATORY FISH OVER THREE YEARS

Time (Months)	Estimated Number of Parrotfish	Estimated Number of Predatory Fish
January 2007	2 500	35
June 2007	2 000	40
December 2007	1 200	55
June 2008	1 100	80
December 2008	970	95
June 2009	400	105
December 2009	200	120

(i) Represent the data in Table 1 on a suitable graph on the grid provided on page 7. **(6 marks)**

(ii) Suggest an explanation for the differences in the shapes of the graphs.

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(2 marks)

(iii) Name TWO features that the predatory fish may have that allow it to carry out its role as a predator.

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(2 marks)

(iv) Suggest what the likely consequences would be if the population size of the predatory fish is NOT controlled.

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(2 marks)

Total 25 marks

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2. (a) Figure 4 shows an external view of the human heart.

(i) Identify the structures labelled A, B and C in Figure 4. **Write your answers on the lines provided in Figure 4**. **(3 marks)**

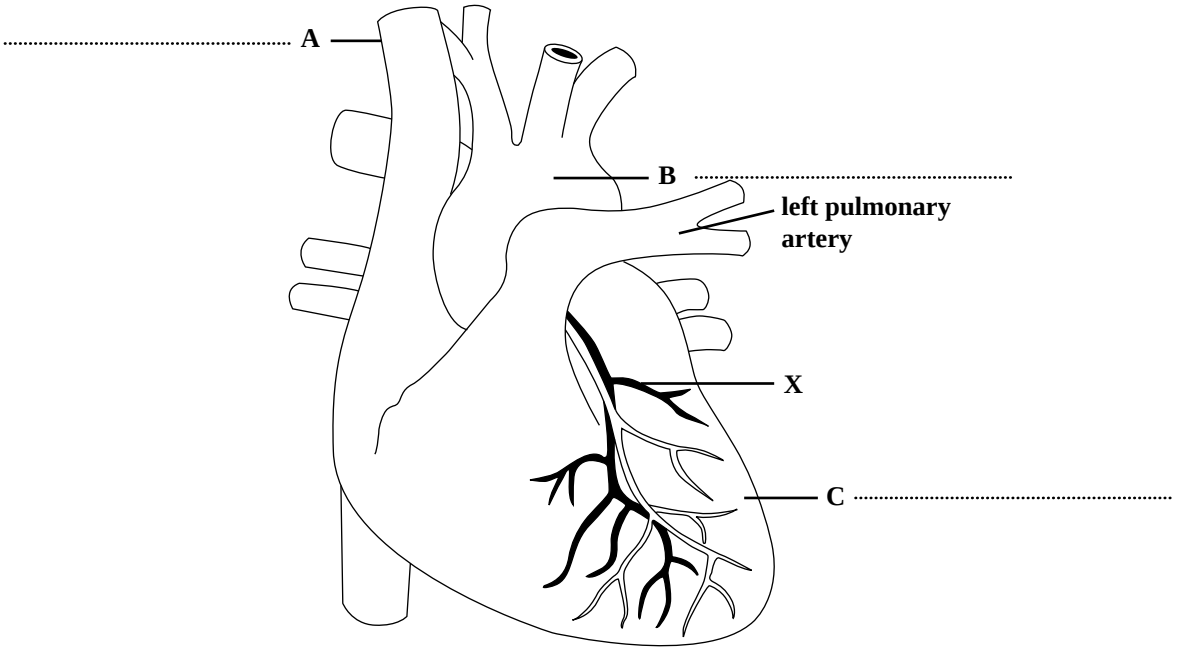

Figure 4. External view of the human heart

(ii) Explain how the function of the heart may be affected by a blockage of the blood vessel shown at X.

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(2 marks)

(b) Figure 5 shows blood vessels used to transport fluids around the body of a mammal.

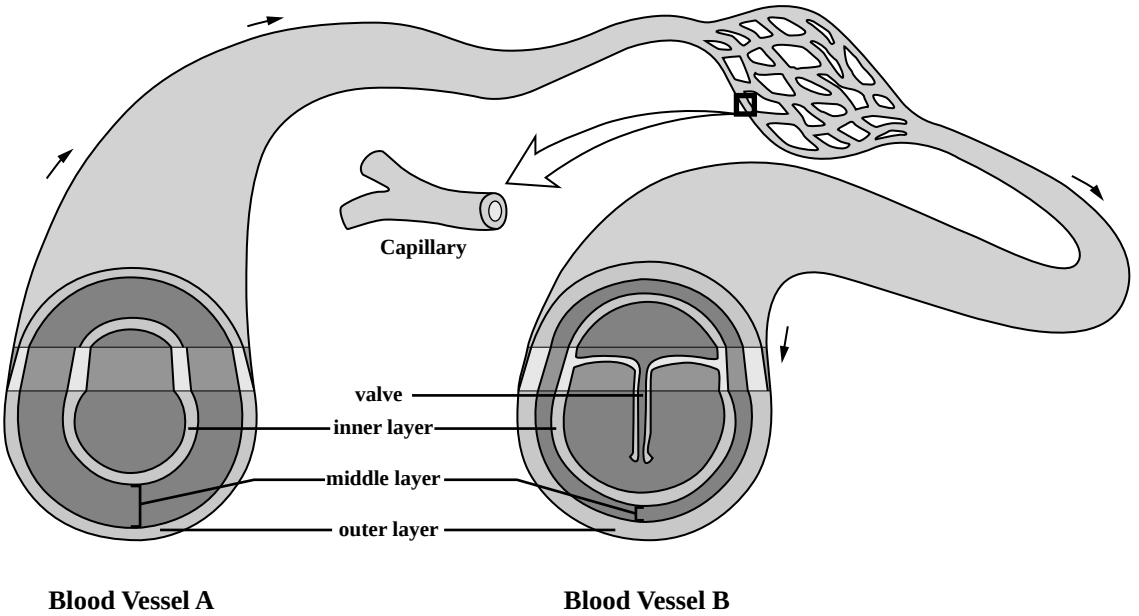

Figure 5. Blood vessels in a mammal

(i) Which blood vessel, A or B, represents a vein?

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(1 mark)

(ii) Identify ONE structure visible in Figure 5 which helps a **vein** to carry out its function.

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(1 mark)

(iii) Explain how the structure identified in (b) (ii) allows the vein to fulfil its function.

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(2 marks)

(iv) Diets high in saturated fats may lead to the formation of plaques in blood vessels. Explain how such deposits may lead to hypertension.

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(2 marks)

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(c) Figure 6 shows the transport vessels of a flowering plant.

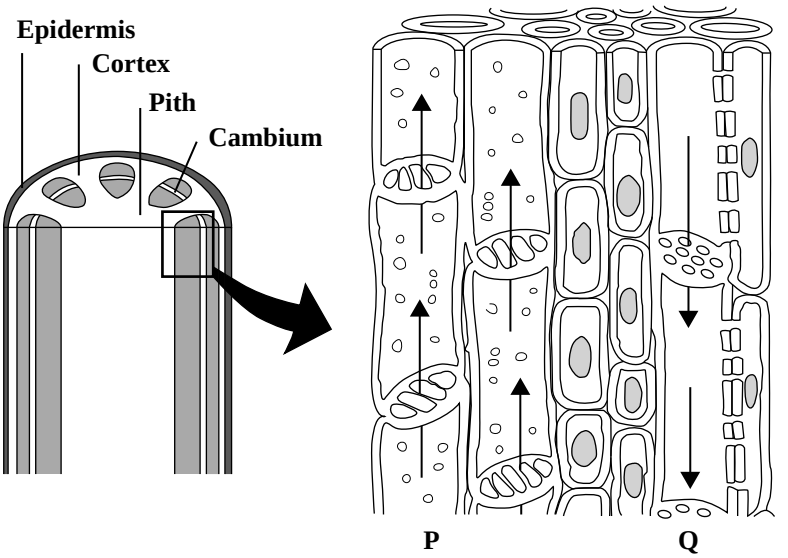

Figure 6. Transport vessels in a flowering plant

Identify the vessels shown at Positions P and Q in Figure 6.

P .....

Q .....

(2 marks)

(d) Explain why plants do NOT need an organ like a heart in their transport system, and outline ONE mechanism by which plants move fluids through their transport vessels.

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(2 marks)

Total 15 marks

3. (a) Figure 7 is a diagram of a nephron found in a human kidney.

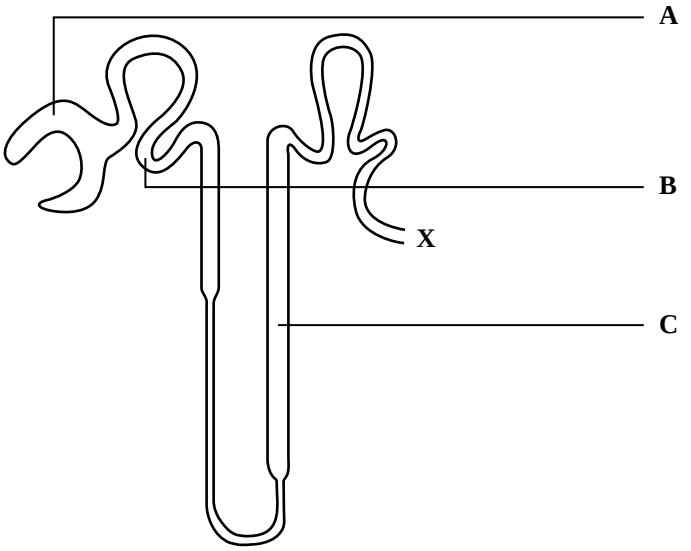

Figure 7. Diagram of a nephron

(i) Identify the structures labelled A, B and C in Figure 7 and briefly describe the process that takes place in EACH of these structures.

A .....

Process .....

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B .....

Process .....

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C .....

Process .....

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(6 marks)

(ii) Name the hormone secreted by the pituitary gland when there is NOT enough water in body cells.

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(1 mark)

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(iii) Explain why the composition of urine produced at X would differ in someone who drank a lot of water compared to someone who had no water to drink for over 24 hours.

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(4 marks)

(b) Table 2 shows a comparison of the substances found in the urine of a healthy person and that of someone who is diabetic.

TABLE 2: COMPOSITION OF URINE OF A HEALTHY PERSON AND OF A DIABETIC PERSON

Substance	% in Urine of Healthy Person	% in Urine of Diabetic Person
Glucose	0	10.0
Protein	0	0
Urea	2.0	6.0
Sodium chloride	0.6	0.4
Water	95	90

(i) Explain why there is glucose in the urine of the diabetic person, assuming that the kidneys are functioning properly.

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(2 marks)

(ii) Describe TWO ways of managing diabetes mellitus.

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(2 marks)

Total 15 marks

SECTION B

Answer ALL questions.

Write your answers in the spaces provided at the end of each question.

4. (a) With the aid of a diagram, outline THREE processes which return carbon to the atmosphere and ONE which removes it. (Include the form in which the carbon is found in the atmosphere.) **(6 marks)**

(b) Describe the impact of **cutting down** or **burning** large areas of forest on the recycling of carbon. **(6 marks)**

(c) Suggest THREE ways in which humans can make changes in their day-to-day activities to reduce the amount of carbon in the atmosphere. **(3 marks)**

Total 15 marks

Write your answer to Question 4 here.

Space for diagram for (a)

Write your answer to Question 4 here.

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Write your answer to Question 4 here.

5. (a) Define the terms ‘genotype’, ‘phenotype’, ‘recessive’ and ‘dominant’. **(4 marks)**

(b) (i) Albinism (lack of pigmentation) is an example of discontinuous variation. Give TWO examples of traits that show **continuous** variation. **(2 marks)**

(ii) A man with normal pigmentation and an albino woman produce two children, one of whom is an albino. Using appropriate symbols, draw a genetics diagram to show the genotypes of the parents and the expected phenotypes of the offspring. (Note that albinism is a trait caused by the inheritance of a recessive allele.) **(5 marks)**

(c) Many of the antibiotics used to control tuberculosis are no longer effective. Use the theory of natural selection to account for this. **(4 marks)**

Total 15 marks

Write your answer to Question 5 here.

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Write your answer to Question 5 here.

Write your answer to Question 5 here.

6. (a) (i) Name THREE types of tissue which coordinate to bring about movement in animals, and state THREE reasons why it is important for animals to be able to move. **(6 marks)**

(ii) Explain why, unlike animals, plants are able to survive without moving from place to place. **(2 marks)**

(b) A father **saw** his son step down onto the tarmac and **watched** as he ran towards him. Describe the changes that occurred in the father's eyes to keep his son's face in focus. **(7 marks)**

Total 15 marks

Write your answer to Question 6 here.

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END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.